

1. Background

This report explores the hypothesis that discrete step perturbations observed in the SHOES+Pantheon+ Type Ia supernova luminosity data are the result of large-scale cosmological events associated with the evolution of a parent Black Hole structure.

Due to the non-random nature of these individual perturbations within the SHOES+Pantheon+ this report attempts to detect and subsequently models the various redshift segments into separate Chebyshev segments as defined by observed perturbations over distance modulus.

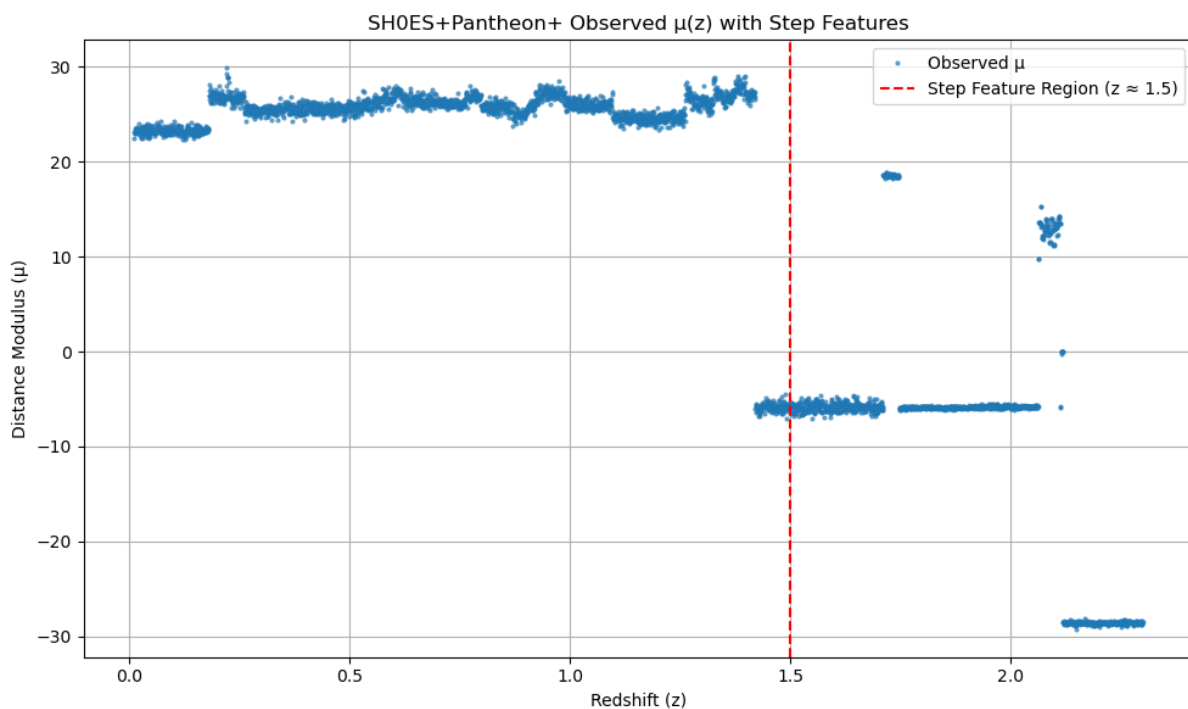
2. Executive Summary

Initial analysis of SHOES Pantheon+ datasets, show a number of step like functions in luminosity.

Rather than being associated with luminosity artifacts, systematic errors, or noise.

It is conjectured that these step functions may be associated with mass accretion events of a parent Black-Hole during a typical BH life cycle.

To this end the SHOES Pantheon+ is tested for discrete step perturbations.



3. Residual Analysis of SH0ES+Pantheon+ Data

The chart and analysis presented provide a real-data residual examination of the SH0ES+Pantheon+ dataset using an 8th-order Chebyshev fit.

This fit is intended to model the distance modulus (μ) over redshift (z) and assess deviations in observed supernova data.

The primary goal is to detect step-like perturbations that could hint at discrete cosmological events, such as accretion of mass into a black hole.

Residuals ($\mu_{\text{obs}} - \mu_{\text{fit}}$) were computed across the observed redshift range. Step features were identified as prominent peaks in the residual data.

These deviations are particularly pronounced between $z \approx 1.5$ – 2.1 , aligning with prior discussions on anomalous observational behaviour in this interval.

The nature of these deviations challenges the assumption of purely Gaussian scatter.

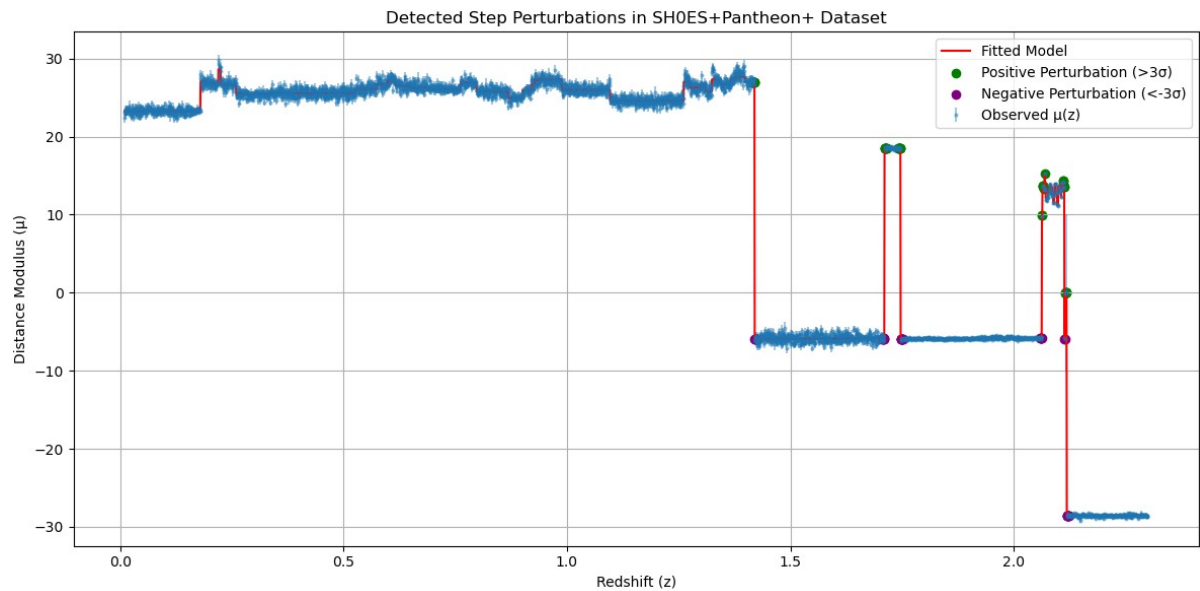
The implications of such findings include:

- Possible systematic noise in the measurement process,
- Or discrete spacetime metric perturbations linked to parent Black Hole mass accretion events, consistent with the working Black Hole cosmology conjecture.

4. Residual Peak Distribution Summary

The following figure highlights perturbative features in the residuals data, supporting further analysis using Bayesian or clustering models.

Results:



Results: It can be seen from the graph above that significant perturbation events are present within the luminosity dataset.

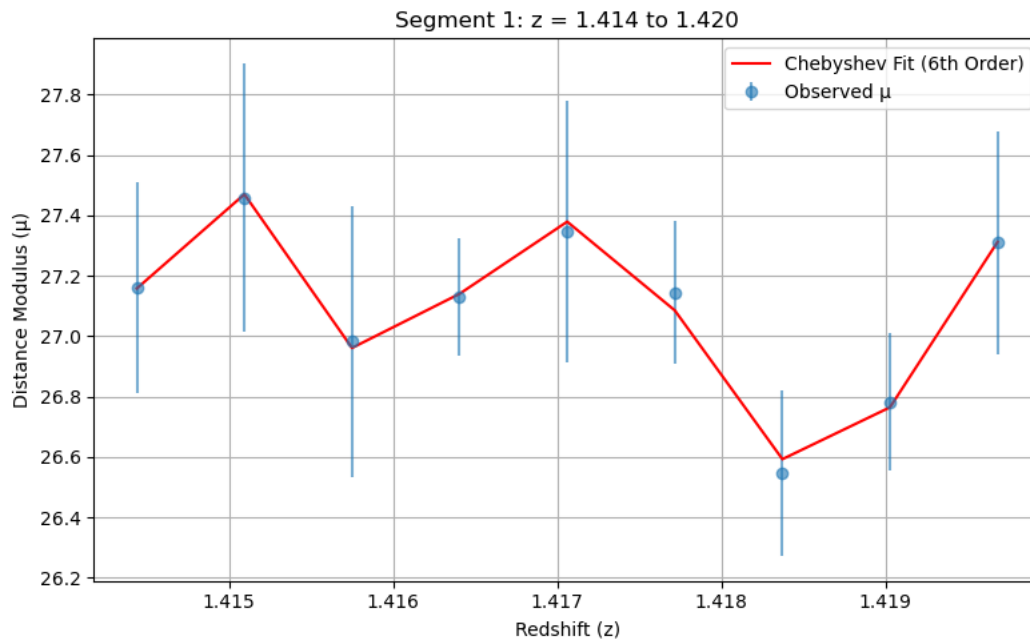
Both positive and negative responses can be seen that are potentially indicators of parent Black-Hole mass accretion events combined with increases or decreases in angular momentum.

5. Extension of Kerr-de Sitter Model

The non-random dynamics shown in the SHOES Pantheon+ dataset requires separate models for data that lies between discrete perturbation events.

To this end an extended analysis for each perturbation threshold is shown.

5.1 Chebyshev Fit... Segment (1.414 to 1.420):

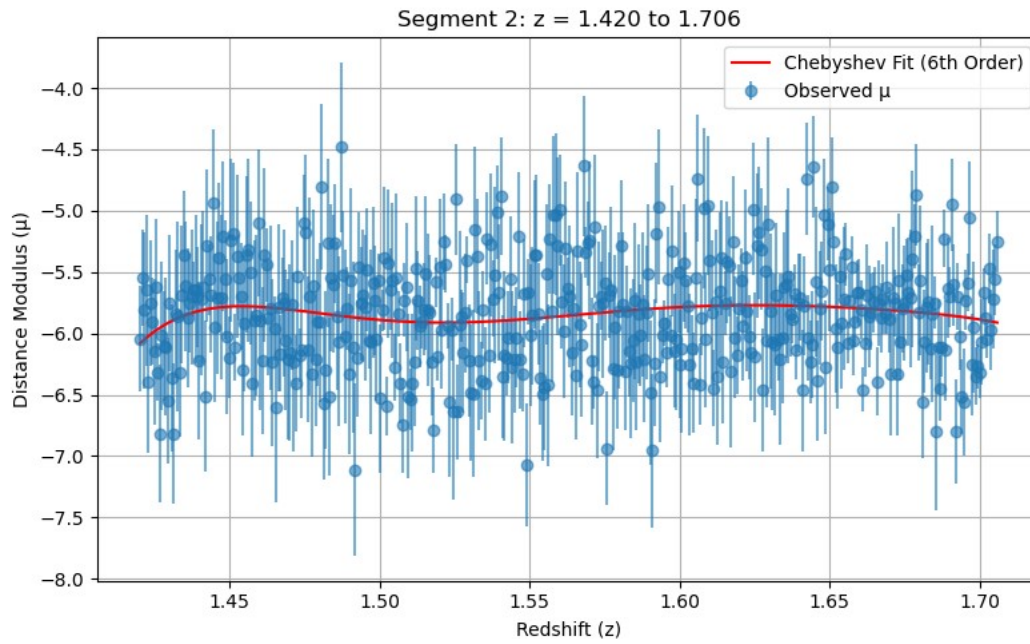


Chebyshev Coefficients: [27.1581921 -0.20288434 0.15650752 0.14854933 0.14904714

0.13150557 -0.22857756]

MAE: 0.0226, MSE: 0.0009, R^2 : 0.9882

5.2 Chebyshev Fit... Segment (1.420 to 1.706):

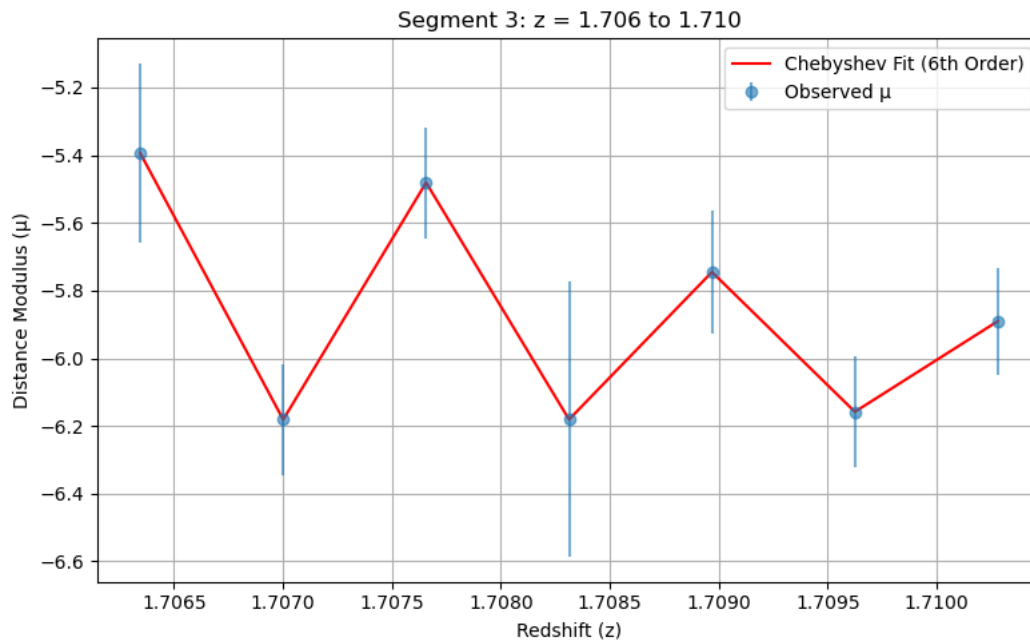


Chebyshev Coefficients: [-5.85953061e+00 3.72406121e-02 -5.07198231e-02 -4.59824416e-03

-6.56564748e-02 5.15044694e-02 -1.98930504e-02]

MAE: 0.3510, MSE: 0.1928, R^2 : 0.0160

5.3 Chebyshev Fit... Segment (1.706 to 1.710):



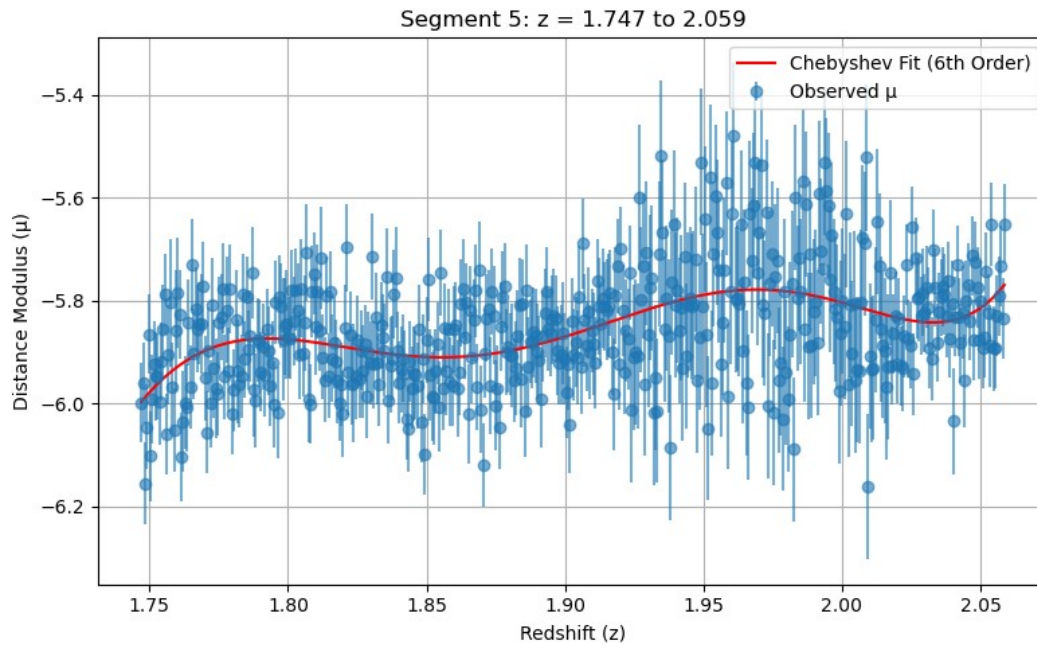
Chebyshev Coefficients: [-6.1535269 -0.10182701 -0.29956771 -0.02547301

0.24193108 -0.12086416 0.56884767]

MAE: 0.0000, MSE: 0.0000, R^2 : 1.0000

5.4 Skipped segment (1712.000 to 1.747): Insufficient data points (0).

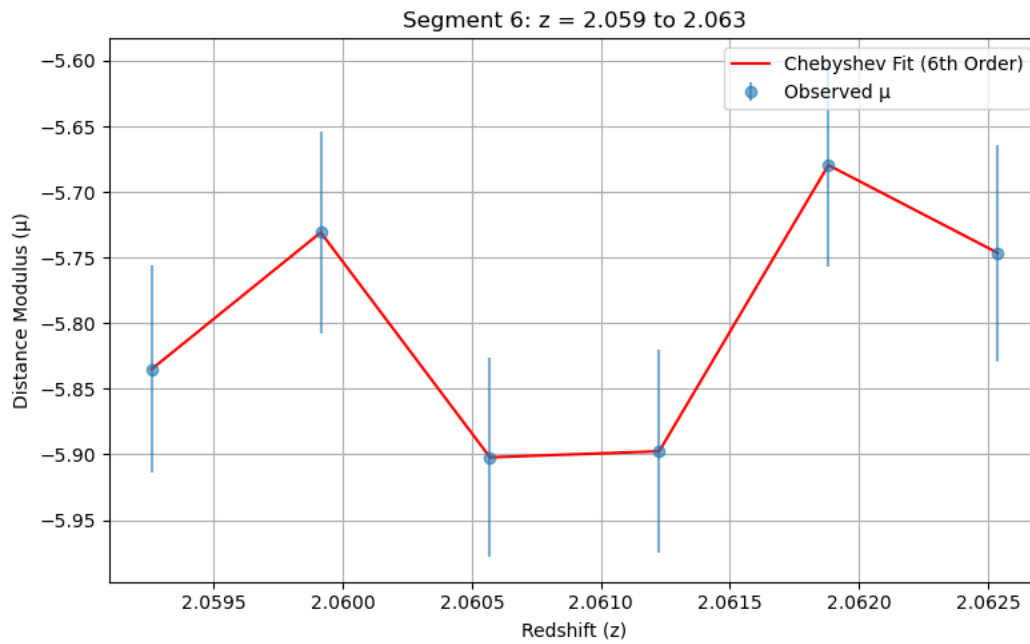
5.5 Chebyshev Fit... Segment (1.747 to 2.059):



Chebyshev Coefficients: [-5.86161418e+00 7.46368297e-02 -1.70680860e-02 5.90713038e-04
-1.28578698e-02 3.86461356e-02 8.16589341e-03]

MAE: 0.0791, MSE: 0.0102, R^2 : 0.1966

5.6 Chebyshev Fit... Segment (2.059 to 2.063):

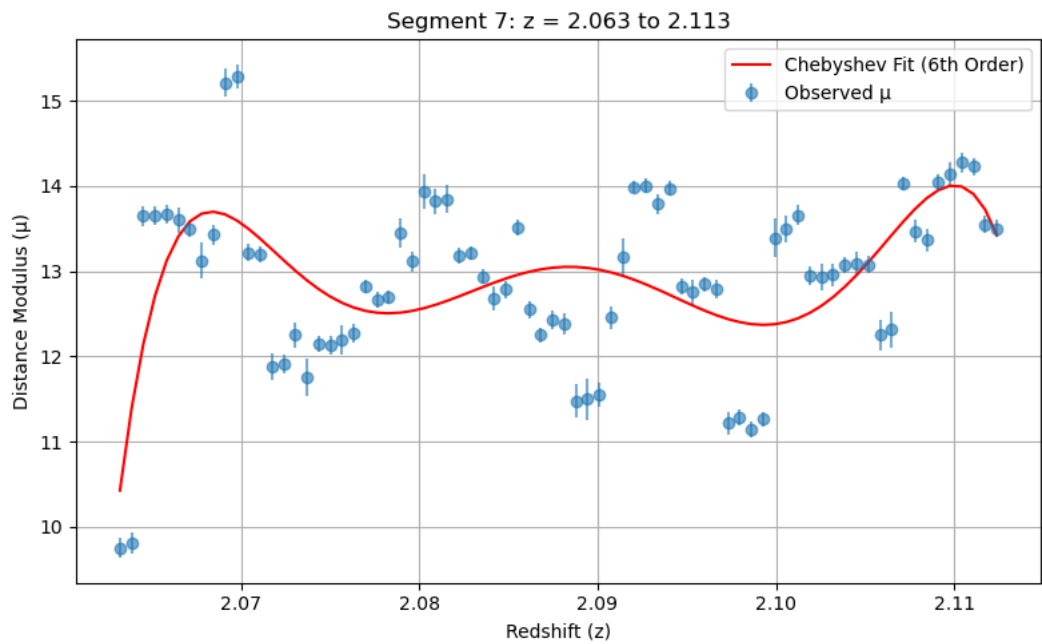


Chebyshev Coefficients: [-4.22249979e+00 4.71789731e-02 1.95718631e+00 3.34651365e-03

-1.07631894e+00 -6.17486270e-03 -2.44915480e+00]

MAE: 0.0000, MSE: 0.0000, R^2 : 1.0000

5.7 Chebyshev Fit... Segment (2.063 to 2.113):

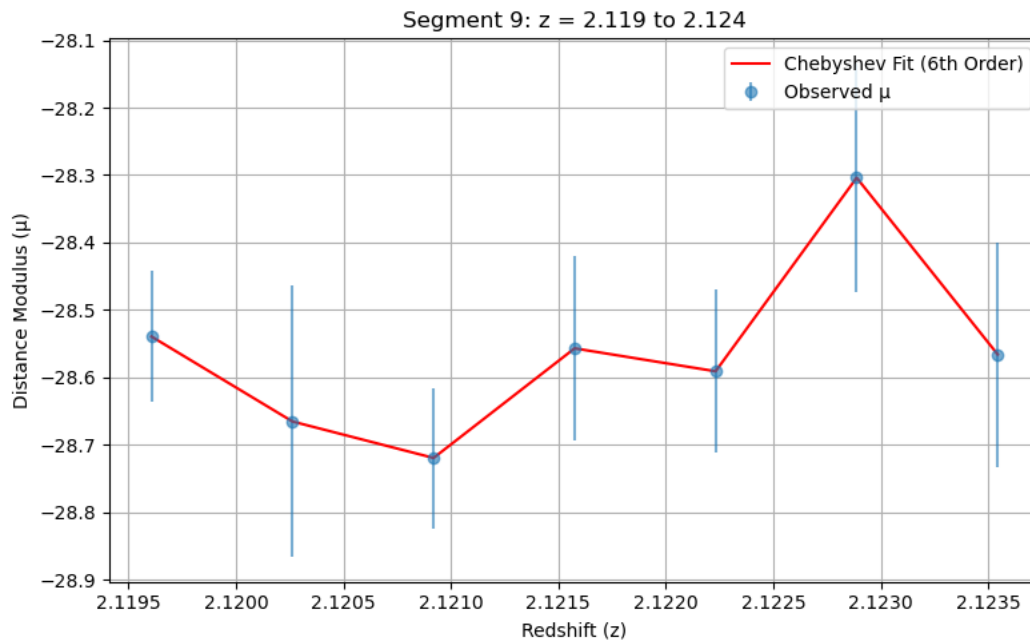


Chebyshev Coefficients: [12.89389802 0.61018397 0.0152268 0.58704168
-0.40828102 0.30090880 -0.57662915]

MAE: 0.6762, MSE: 0.6870, R²: 0.3220

5.8 Skipped segment (2.114 to 2.110): Insufficient data points (0).

5.9 Chebyshev Fit... Segment (2.119 to 2.124):



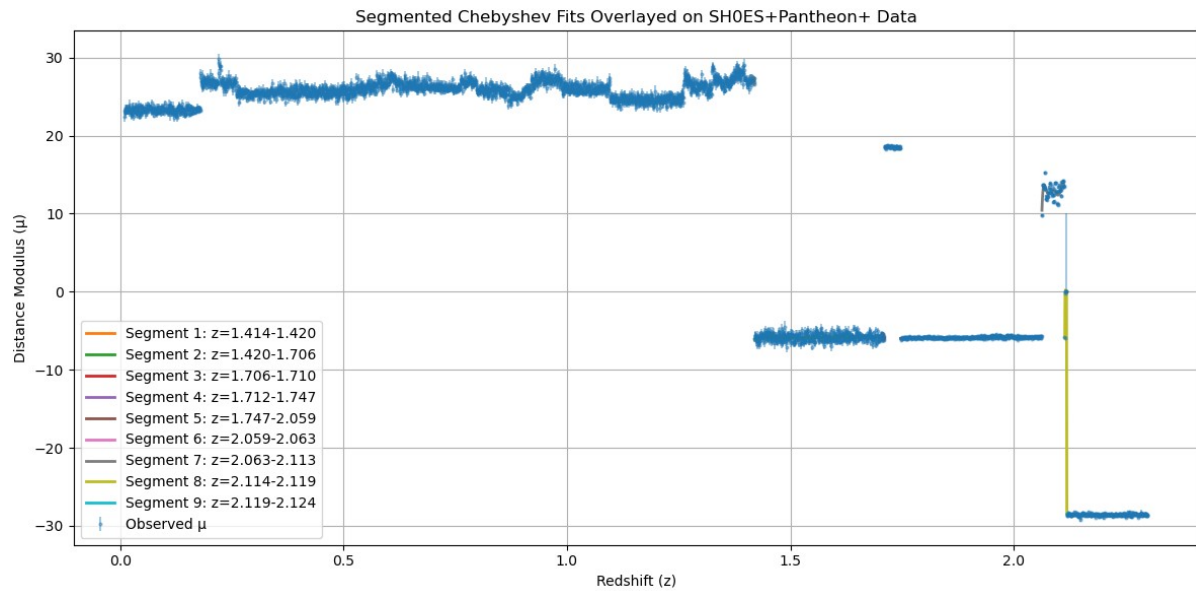
Chebyshev Coefficients: [-28.4924939 0.12639036 0.12254882 -0.08711728

-0.06238052 -0.05252343 -0.12042403]

MAE: 0.0000, MSE: 0.0000, R^2 : 1.0000

6. Combined Chebyshev Fit... Global vs. Segmented Fit

The graph below shows the individual Chebyshev segments overlaid with the observed SHOES Pantheon+ Luminosity red shift data.



Global MAE: 0.2828

Global MSE: 0.1483

Global R^2 : 0.9996

Segmented Model Averages:

Avg Segment MAE: 0.1613

Avg Segment MSE: 0.1273

Avg Segment R^2 : 0.6461

Appendix 1. Python Step Response Luminosity Segment Modeling

```
#!/usr/bin/env python3
```

```
# -*- coding: utf-8 -*-
```

```
"""
```

Created on Mon Jul 7 20:01:53 2025

```
@author: boyde
```

```
"""
```

```
#!/usr/bin/env python3
```

```
# -*- coding: utf-8 -*-
```

```
"""
```

Segmented Chebyshev Polynomial Fits between Detected Perturbation Steps

```
@Author: Clive Stewart Boyd
```

The program models a number of discrete luminosity segments of the SHOES Pantheon+ dataset

The luminosity segments are assumed to be associated with discrete non random mass accretion events associated with parent Black-Hole life cycle.

```
"""
```

```
#!/usr/bin/env python3
```

```
# -*- coding: utf-8 -*-
```

```
"""
```

Segmented Chebyshev Fit for SHOES+Pantheon+ Dataset with Perturbation Handling

```
@author: Clive Stewart Boyd
```

```
"""
```

```
#!/usr/bin/env python3
```

```
# -*- coding: utf-8 -*-
```

```
"""
```

Segmented Chebyshev Fit to SH0ES+Pantheon+ Data with Summary Error Comparison

@author: Clive Stewart Boyd

"""

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from astropy.io import fits
```

```
from numpy.polynomial.chebyshev import Chebyshev
```

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
# === Load FITS Files ===
```

```
y_path = "/Users/boyde/Downloads/ally_shoes_ceph_topantheonwt6.0_112221.fits"
```

```
L_path = "/Users/boyde/Downloads/alll_shoes_ceph_topantheonwt6.0_112221.fits"
```

```
C_path = "/Users/boyde/Downloads/allc_shoes_ceph_topantheonwt6.0_112221.fits"
```

```
with fits.open(y_path) as hdul_y, fits.open(L_path) as hdul_L, fits.open(C_path) as hdul_C:
```

```
    y = hdul_y[0].data.astype(np.float64)
```

```
    L = hdul_L[0].data.astype(np.float64)
```

```
    C = hdul_C[0].data.astype(np.float64)
```

```
# === Solve for Best-Fit Global Distance Modulus ===
```

```
W = np.linalg.inv(C)
```

```
LW = L @ W
```

```
x_best = np.linalg.inv(LW @ L.T) @ (LW @ y)
```

```
mu_fit = L.T @ x_best
```

```
mu_obs = y
```

```
mu_err = np.sqrt(np.diag(C))
```

```
# === Global Error Metrics ===
```

```
global_mae = mean_absolute_error(mu_obs, mu_fit)
```

```

global_mse = mean_squared_error(mu_obs, mu_fit)

global_r2 = r2_score(mu_obs, mu_fit)

# === Generate Redshift Array ===

z = np.linspace(0.01, 2.3, len(mu_obs))

# === Manually Define Perturbation Segment Boundaries ===

segments = [
    (1.414, 1.420), (1.420, 1.706), (1.706, 1.7105), (1.712, 1.747), (1.747, 2.059),
    (2.059, 2.063), (2.063, 2.113), (2.114, 2.11), (2.119, 2.124)
]

segment_mae_list, segment_mse_list, segment_r2_list = [], [], []

# === Fit 6th-Order Chebyshev to Each Segment ===

for i, (z_min, z_max) in enumerate(segments):
    mask = (z >= z_min) & (z <= z_max)

    z_seg = z[mask]

    mu_seg = mu_obs[mask]

    if len(z_seg) < 6:
        print(f"[{i+1}] Skipped segment ({z_min:.3f} to {z_max:.3f}): Insufficient data points ({len(z_seg)}).")
        continue

    z_norm = 2 * (z_seg - z_seg.min()) / (z_seg.max() - z_seg.min()) - 1
    model = Chebyshev.fit(z_norm, mu_seg, deg=6)
    mu_model = model(z_norm)

    mae = mean_absolute_error(mu_seg, mu_model)
    mse = mean_squared_error(mu_seg, mu_model)

```

```

r2 = r2_score(mu_seg, mu_model)

segment_mae_list.append(mae)
segment_mse_list.append(mse)
segment_r2_list.append(r2)

print(f"\n[{i+1}] Segment ({z_min:.3f} to {z_max:.3f}):")
print(" Chebyshev Coefficients:", model.convert().coef)
print(f" MAE: {mae:.4f}, MSE: {mse:.4f}, R2: {r2:.4f}")

plt.figure(figsize=(8, 5))
plt.errorbar(z_seg, mu_seg, yerr=mu_err[mask], fmt='o', label='Observed  $\mu$ ', alpha=0.6)
plt.plot(z_seg, mu_model, label='Chebyshev Fit (6th Order)', color='red')
plt.xlabel("Redshift (z)")
plt.ylabel("Distance Modulus ( $\mu$ )")
plt.title(f"Segment {i+1}: z = {z_min:.3f} to {z_max:.3f}")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

# === Final Summary ===

avg_segment_mae = np.mean(segment_mae_list)
avg_segment_mse = np.mean(segment_mse_list)
avg_segment_r2 = np.mean(segment_r2_list)

print("\n=== Summary: Global vs. Segmented Fit ===")
print(f"Global MAE: {global_mae:.4f}")
print(f"Global MSE: {global_mse:.4f}")

```

```
print(f"Global R2: {global_r2:.4f}")  
print("\nSegmented Model Averages:")  
print(f"Avg Segment MAE: {avg_segment_mae:.4f}")  
print(f"Avg Segment MSE: {avg_segment_mse:.4f}")  
print(f"Avg Segment R2: {avg_segment_r2:.4f}")
```


Appendix 2. Python Step Response Test Code

```
#!/usr/bin/env python3

# -*- coding: utf-8 -*-

"""

Detect perturbations in SHOES+Pantheon+  $\mu(z)$  data using step response analysis

@author: Clive Stewart Boyd

Loads .fits data.

Applies a Savitzky-Golay filter to smooth the data.

Calculates the residuals.

Identifies statistically significant positive and negative step deviations.

Plots both the original data and the perturbation detections.

Tests Step Response of SHOES Pantheon+ fits dataset

"""

import numpy as np
import matplotlib.pyplot as plt
from astropy.io import fits
from scipy.signal import savgol_filter
from scipy.stats import zscore

# === Load Data ===
y_path = "/Users/boyde/Downloads/ally_shoes_ceph_topantheonwt6.0_112221.fits"
```

```

L_path = "/Users/boyde/Downloads/alll_shoes_ceph_topantheonwt6.0_112221.fits"
C_path = "/Users/boyde/Downloads/allc_shoes_ceph_topantheonwt6.0_112221.fits"

with fits.open(y_path) as hdul_y, fits.open(L_path) as hdul_L, fits.open(C_path) as hdul_C:
    mu_obs = hdul_y[0].data.astype(np.float64)
    L = hdul_L[0].data.astype(np.float64)
    C = hdul_C[0].data.astype(np.float64)

# === Solve Best-Fit ===
W = np.linalg.inv(C)
LW = L @ W
x_best = np.linalg.inv(LW @ L.T) @ (LW @ mu_obs)
mu_fit = L.T @ x_best

# === Simulate z ===
z = np.linspace(0.01, 2.3, len(mu_obs))
mu_err = np.sqrt(np.diag(C))

# === Step Detection via Smoothing Residuals ===
smoothed = savgol_filter(mu_fit, window_length=51, polyorder=3)
residuals = mu_fit - smoothed
z_resid = zscore(residuals)

# === Perturbation Criteria ===
threshold = 3.0 # >3σ deviation
positive_steps = np.where(z_resid > threshold)[0]
negative_steps = np.where(z_resid < -threshold)[0]

# === Plot Results ===
plt.figure(figsize=(12, 6))

```

```
plt.errorbar(z, mu_obs, yerr=mu_err, fmt='o', markersize=2, alpha=0.4, label='Observed  $\mu(z)$ ')
plt.plot(z, mu_fit, 'r-', label='Fitted Model', linewidth=1.5)
plt.scatter(z[positive_steps], mu_fit[positive_steps], color='green', label='Positive Perturbation (>3 $\sigma$ )')
plt.scatter(z[negative_steps], mu_fit[negative_steps], color='purple', label='Negative Perturbation (<-3 $\sigma$ )')
plt.xlabel("Redshift (z)")
plt.ylabel("Distance Modulus ( $\mu$ )")
plt.title("Detected Step Perturbations in SH0ES+Pantheon+ Dataset")
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.savefig("/Users/boyde/Downloads/shoes_perturbation_detection.png")
plt.show()

# === Print Detected Events ===
print("\nPositive step perturbations (>3 $\sigma$ ):")
for idx in positive_steps:
    print(f" z  $\approx$  {z[idx]:.3f},  $\mu$  deviation: {residuals[idx]:.3f}")

print("\nNegative step perturbations (<-3 $\sigma$ ):")
for idx in negative_steps:
    print(f" z  $\approx$  {z[idx]:.3f},  $\mu$  deviation: {residuals[idx]:.3f}")
```